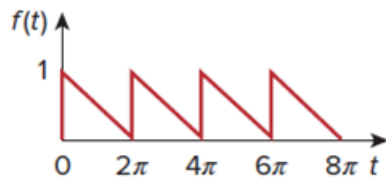


Problem

Obtain the Laplace transform of the periodic waveform in Fig. 15.33.

Figure 15.33:



Step-by-step solution

Step 1 of 1

$$T = 2\pi$$

$$\text{Let } f_1(t) = \left(1 - \frac{t}{2\pi}\right) [u(t) - u(t - 2\pi)]$$

$$f_1(t) = u(t) - \frac{t}{2\pi} u(t) + \frac{1}{2\pi} (t - 2\pi) u(t - 2\pi)$$

$$F_1(s) = \frac{1}{s} - \frac{1}{2\pi s^2} + \frac{e^{-2\pi s}}{2\pi s^2} = \frac{2\pi s + [-1 + e^{-2\pi s}]}{2\pi s^2}$$

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}} = \frac{2\pi s - 1 + e^{-2\pi s}}{2\pi s^2 (1 - e^{-2\pi s})}$$